1 Requirements Gathering

1.1 Stakeholder Analysis

The Used Car Price Prediction System involves several stakeholders who benefit from accurate and reliable vehicle price estimation. Each stakeholder has different expectations and requirements from the system.

The system is designed to satisfy these stakeholders by providing accurate predictions using machine learning models trained on vehicle specifications such as mileage, year, fuel type, engine type, transmission, and brand.

1.2 User Stories and Use Cases

1.2.1 User Stories

Story 1 – Car Buyer As a car buyer, I want to enter vehicle specifications and receive an estimated market price so that I can determine whether the car is overpriced or fairly priced.

Story 2 – Car Seller As a car seller, I want to estimate the value of my car before listing it for sale so that I can set a competitive and realistic price.

Story 3 – Car Dealer As a car dealer, I want to evaluate multiple vehicles quickly so that I can make informed purchasing and selling decisions.

Story 4 – Administrator As a system administrator, I want the trained models and preprocessing pipeline to be saved and reusable so that the system can perform predictions consistently.

1.2.2 Main Use Case: Predict Car Price

1.2.3 Secondary Use Case: Compare Models

The project compares several regression models including Linear Regression, Ridge Regression, Lasso Regression, Elastic Net Regression, Decision Tree Regressor, and Gradient Boosting Regressor.

1.3 Functional Requirements

The system must provide the following functionalities:

1.3.1 Data Management

- The system shall load and process large automotive datasets.
- The system shall handle missing values and remove duplicates.
- The system shall standardize column names and clean raw data.

1.3.2 Data Analysis and Preprocessing

- The system shall perform exploratory data analysis (EDA).
- The system shall detect and handle outliers using IQR and Winsorization techniques.

Stakeholder	**Role**	**Needs and Expectations**
Car Buyers	Individuals searching for fair vehicle prices	Accurate price estimation, easy-to-use interface, reliable predictions
Car Sellers	Owners selling used vehicles	Fair market valuation, quick price estimation, transparency
Car Dealers	Businesses trading used cars	Bulk price prediction, fast processing, high prediction accuracy
System Administrators	Maintain and manage the system	System stability, scalability, secure model storage
Data Scientists / Developers	Build and improve the ML models	Clean datasets, model evaluation metrics, retraining capability
Automotive Businesses	Integrate prediction services into platforms	API compatibility, deployment readiness, real-time response

Table 1: Stakeholder Analysis

Element	**Description**
Use Case Name	Predict Used Car Price
Primary Actor	Buyer / Seller / Dealer
Preconditions	User provides valid vehicle information
Input	Year, mileage, brand, model, engine type, transmission, power, etc.
Process	System preprocesses input, applies encoding and scaling, then predicts the car price using the trained machine learning model
Output	Estimated used car price
Postconditions	Predicted price displayed to user

Table 2: Main Use Case

Element	**Description**
Use Case Name	Compare Machine Learning Models
Primary Actor	Developer / Data Scientist
Preconditions	Dataset is cleaned and prepared
Process	Train multiple regression models and evaluate them using R^2, RMSE, and MAE metrics
Output	Best-performing model selected
Postconditions	Best model saved for deployment

Table 3: Model Comparison Use Case

1.3.5 Deployment and Reusability

- The system shall save preprocessing components including encoders and scalers.
- The system shall support deployment using Flask or FastAPI.

1.4 Non-Functional Requirements

1.4.1 Performance Requirements

- The system should generate predictions within a few seconds.
- The system should efficiently process datasets containing hundreds of thousands of records.
- The system should support parallel computation during training when possible.

1.4.2 Security Requirements

- Saved models and preprocessing files should be securely stored.
- Input validation should prevent invalid or corrupted user data.
- Only authorized administrators should modify training datasets and saved models.

1.4.3 Usability Requirements

- The interface should be simple and user-friendly.
- Users should easily input vehicle specifications and receive predictions.
- Results should be displayed clearly and understandably.

1.4.4 Reliability Requirements

- The system should produce consistent predictions for identical inputs.
- Preprocessing steps must remain identical during both training and inference.
- Saved pipelines and models should ensure reproducibility.

1.4.5 Scalability Requirements

- The system should support adding new machine learning models in the future.
- The architecture should allow integration with web applications and APIs.
- Future enhancements such as XGBoost and Neural Networks should be supported.